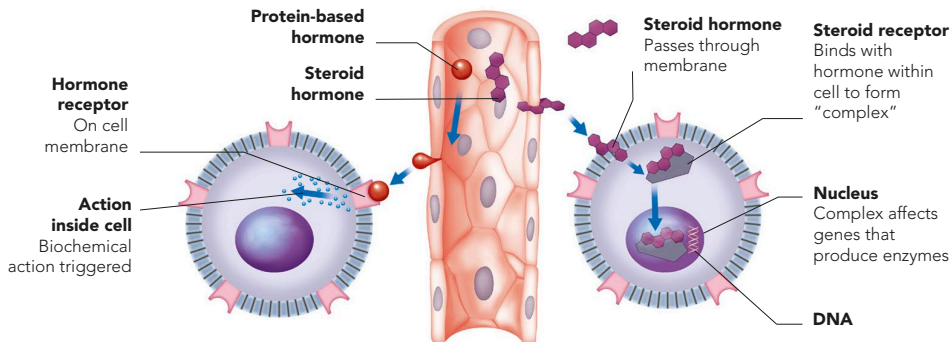


HORMONE CONTROL MECHANISMS

Chemically, there are two main types of hormones: those made of protein and amine molecules, and those made of steroids. The two groups work in a similar way, biochemically altering production

rates of certain substances, but at a cellular level they have different mechanisms. Protein and amine hormones act on receptor sites at a cell's surface; steroid hormones act on receptors inside the cell.



PROTEIN-BASED HORMONES

These hormones cannot pass through the cell membrane. They bind to receptors on the membrane, triggering biochemical action inside the cell.

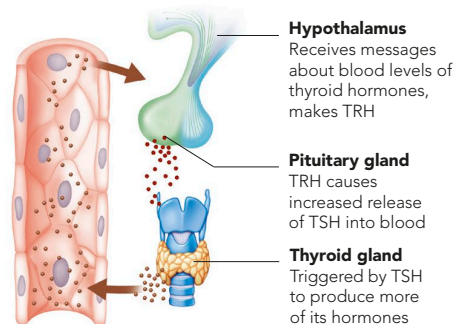
STEROID-BASED HORMONES

Steroids pass into the cell, then bind to receptors and enter the cell nucleus. This triggers genes to produce enzymes that prompt biochemical action.

FEEDBACK MECHANISMS

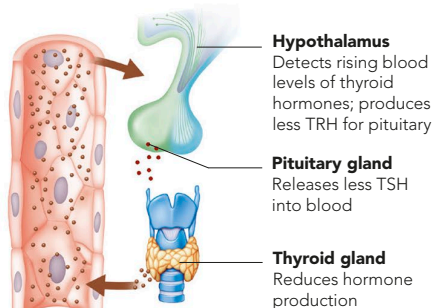
Hormone levels are controlled by feedback mechanisms, or loops. The amount of a hormone in the blood is detected and passed on to a control unit, which in many cases is the hypothalamus–pituitary

complex (as with the thyroid hormones, see below). If a hormone level is too high, the control unit reduces hormone production. If the level is too low, the control unit stimulates production.



INCREASING LEVELS

Thyrotropin-releasing hormone (TRH) from the hypothalamus causes the pituitary to make thyroid-stimulating hormone (TSH); hormone levels rise.



DECREASING LEVELS

High hormone levels prompt negative feedback, so the hypothalamus produces less TRH. This reduces TSH levels and the thyroid produces fewer hormones.